

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

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Finding Reasoning Paths :

1. **Anatomical Snuffbox**: The anatomical snuffbox is a triangular depression on the radial side of the hand , bounded by the extensor pollicis longus tendon laterally and the tendons of the extensor pollicis brevis and abductor pollicis longus medially . It contains the **radial artery** .
2. **Penetrating Wound**: A penetrating stab wound in this region is likely to involve the **radial artery** , which runs superficially in this area .
3. **Excessive Bleeding**: The radial artery is a major artery supplying the hand and forearm , and its location in the anatomical snuffbox makes it highly vulnerable to penetrating injuries , leading to significant bleeding .
4. **Vascular Supply**: The radial artery is a branch of the **brachial artery** and is responsible for supplying blood to the hand and forearm .

Reasoning Process :

1. **Anatomical Context**: The anatomical snuffbox is located on the radial side of the hand , and its boundaries are well-defined by tendons of the forearm muscles . The radial artery runs along the floor of this depression , making it highly exposed in this area .
2. **Mechanism of Injury**: A penetrating stab wound to the anatomical snuffbox is likely to damage the **radial artery** because of its superficial location .
3. **Clinical Presentation**: The patient presents with severe bleeding from the wound , which is consistent with damage to a major artery . The anatomical snuffbox is a common site for arterial bleeding due to its anatomical position .
4. **Vascular Anatomy**: The radial artery is a direct branch of the **brachial artery** and is responsible for arterial blood supply to the hand and forearm . Damage to this artery would result in significant bleeding .
5. **Differential Diagnosis**: Other potential causes of bleeding , such as venous injury or injury to other arteries (e.g. , ulnar artery) , are less likely given the location of the wound and the description of the injury .

Conclusion :

The **radial artery** is the most likely source of the excessive bleeding due to its superficial location in the anatomical snuffbox and its role as a major artery supplying the hand and forearm .

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<answer>

Radial artery

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